Fiction That Feeds Back: From Non-Euclidean Geometry to Machine Worldviews

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Abstract

Fiction is not evidence, but it is a recurrent, checkable input to the exact sciences: four channels — formal, controlled, quantitative, narrative — separate under three checks (temporal priority, dispensability of the fictional input, operative residue), and whether machine worldviews open the reverse channel stands or falls on the same falsifiability criteria we specify. This position paper makes that sentence operational. It separates the four channels, each anchored in cases: non-Euclidean geometry as formal language prior to physical reference; thought experiments as controlled fiction under a translatability criterion; entropy as an abstract quantity that entered engineering by formal analogy and re-entered physics carrying a measurable price; and narrative fiction that supplies vocabulary and sets agendas, from a wormhole question to the singularity. A mapping (M1–M5) pairs the hardness criteria of hard science fiction with the falsifiability criteria of AI research claims in a constructed structural parallel — criteria with shared failure modes, one row an import from the pre-registration literature, declared as such — and a four-stage pipeline — setting, formalization, pre-registration, experiment — ports the mapping into a worked method, demonstrated on one artifact at the stated power; one engineering-grade, mid-strength instance is examined with its negative results disclosed in full. The reverse channel is not asserted; an adjudication agenda states what would count as feedback and how it would die. Every claim is stated to be checkable by a third party; for the engineering record, the full frozen record and digests are available from the author on request, public availability deferred to a revised version, and the instance's temporal anchor is an author-attested freeze, recorded as provisional (§7.4).

1 Introduction

1.1 From decoration to input

Fiction, in the standard account of the exact sciences, is what the exact sciences are not: it decorates the work, motivates it, and supplies the contrast against which rigor is defined. This paper takes a narrower and more mechanical view. We ask when a piece of fiction — a formal system built with no intended referent, a thought experiment, a quantity defined over an abstract ensemble, a story — can be shown to have entered the exact sciences as a working input; and, symmetrically, what it would take for such an entry to fail. Treated this way, the subject admits dated anchors, explicit criteria, and verdicts that a third party can re-derive.

Our subject is a two-way loop between fiction and the exact sciences. The forward direction runs from fiction into science, and we argue that it divides into four channel types — formal, controlled, quantitative, narrative — each documented by anchored cases and each scored on the same three checks (§2–§5). The reverse direction — machine worldviews feeding back into

the sciences — is not asserted anywhere in this paper; it is set up as an open question with named criteria (§8), because whether it exists is exactly the kind of claim that can die under a pre-registered adjudication. The business of this paper is criteria, cases, and scores, not celebration. Section 6 supplies the mapping that makes the criteria portable, §7 exhibits them end to end on an engineering artifact, and §9 turns the criteria on the paper itself.

1.2 The claim, its falsifiability conditions, and the three checks

The thesis of this paper is:

Fiction is not evidence, but it is a recurrent, checkable input to the exact sciences: four channels — formal, controlled, quantitative, narrative — separate under three checks (temporal priority, dispensability of the fictional input, operative residue), and whether machine worldviews open the reverse channel stands or falls on the same falsifiability criteria we specify.

Three delimitations are part of the claim, not qualifications bolted onto it. We do not claim that fiction is necessary to science; we do not claim that science descends from fiction; and we do not claim that the reverse channel exists. "Checkable" carries its ordinary methodological sense: the contribution credited to a fictional input must be confirmable or criticizable by named checks, not by taste. "Decidable" is reserved for the pre-registered tier (§6–§7); the thesis sentence deliberately does not use it. On the historical tier (§2–§5), a checkable claim is one a third party can re-examine against dated anchors, and the D check's bounded adjudicative power is recorded rather than concealed (§1.2). The paper claims re-examinability for its historical claims and falsifiability only for its pre-registered ones, and the master table (§9.1) is graded to that split.

The claim is falsifiable in four specified ways.

- **F-C1 (typology).** If the four channel types prove pairwise inseparable under the three checks — if the checks return the same profile for types that are supposed to differ — the four-way typology is weakened. Operational reading: for each pair of channel types, at least one scored flagship case must return a differing grade on at least one of T, D, or R; the condition triggers when some pair returns identical grade profiles across all scored flagship cases of both types.
- **F-C2 (channel).** If the flagship case of any channel fails the temporal check — the dated priority of the fictional input collapses — or cannot sustain even the supply-level claim, leaving no operative residue behind, that channel is downgraded. Operational reading: the temporal check fails when the earliest dated record of the fictional input is not earlier than the earliest dated record of the scientific use credited to it; the supply-level claim fails when no primary record of the development carries an operative residue of the input. A demonstrated dispensable input is not a downgrade trigger under this condition: the channel claims are stated at the supply level, so a fiction-free account of the same development would re-confirm the grading rather than refute the claim.
- **F-C3 (reverse channel).** If hypotheses generated by AI systems, run under pre-registered adjudication, systematically leave no kill line to hang — no falsifiable consequence that a decisive test can reject — then the proposed reverse channel ("machine worldviews feed back into science") is declared dead. Operational reading: systematically means in every case of a sample whose size, source, and counting rule are fixed in the adjudication protocol before the run; a death declared on a post-hoc or moving sample would itself fail the pre-registration it is invoked under.
- **F-C4 (non-collage).** If new cases cannot be classified without ambiguity into one of the four forms under the same three checks, the typology is ad hoc. Operational reading: classification is unambiguous when two classifiers working independently from the case record and the definitions of §1 assign exactly one form each and agree; persistent disagreement, or persistently more than one admissible form, is the ambiguity that triggers the condition.

The three checks are applied to every channel, section by section, and consolidated in §9.1.

- **T (temporal priority).** The fictional input must precede the scientific use it is credited with, and the precedence must be anchorable — by etymology and dictionary record, by primary text, or by participant self-statement.
- **D (dispensability).** An account of the same scientific development that omits the fictional input must be visibly less parsimonious. If an equally parsimonious fiction-free account exists, the channel's contribution is downgraded from necessity to available convenience.
- **R (operative residue).** The encounter must leave an operative residue — a vocabulary, a quantity, an agenda item, or a formal system that continues to do work after the encounter.

The D check returns one of four dispositions, and the dictionary is stated once here so that every later table reads against it. **Partial pass**: the pool contains positive evidence of non-uniqueness, and the claim is downgraded to the supply level (Channel I). **Unresolved by design**: the check pits live positions against each other and the claim is worded to be invariant under either resolution (Channel II). **Indeterminate**: the pool contains no study that could run the check (Channels III and IV). **Out of scope**: the claim never invokes indispensability, so the check has no target (the §7 instance, whose claim rests on executability). The four labels state different reasons the check stops short of a verdict — evidence of non-uniqueness, a live dispute the claim is invariant under, a missing study, or no target at all — and none of them is a verdict of indispensability; no channel in this paper claims one.

Two evidence tiers appear in this paper and are never mixed. Historical evidence is graded by anchor type — primary-source anchor, self-statement anchor, lexical anchor, named textual anchor, quantitative study — and no historical claim is worded to need more than its tier supports. The engineering case of §7 is graded on the pre-registered adjudication tier, a different instrument. Mixing — a claim borrowing the authority of the other tier — never happens; porting, in §6 and §8, is a criteria operation: the checks travel and are re-instantiated at the receiving tier's own grade.

The typology itself carries a positive classification principle, not only a residual one. The channels are separated by the form in which a fictional input enters science — a self-consistent symbolic system (Channel I), a bounded scenario (Channel II), a defined quantity (Channel III), a narrative (Channel IV) — and the four forms are distinguished by mutually exclusive, third-party-checkable objects: a formal system's adoption status, a scenario's reconstructible argument, a quantity's formula and price, a narrative's datable textual trail. The one boundary that could blur — scenario versus narrative — is fixed by a translatability gate: a reconstructible argument makes Channel II, a dated-text-and-reception object makes Channel IV, and translatability takes priority where both readings are available. The D check is declared an asymmetric weak test: it can downgrade, never confirm, and on the current pool it returns no clean verdict — one partial pass, one unresolved by design, two indeterminate, one out of scope — a record, not a defect, of a check whose function is to lower claims to the level their evidence supports. Coverage is not a closed result; F-C4 holds it open to a counterexample channel.

1.3 Contributions and scope

The paper makes five contributions, ordered by what the checks can carry rather than by narrative precedence.

1. **A criterion apparatus for fiction-to-science feedback**: the three checks (temporal priority, dispensability, operative residue) with a declared disposition dictionary for D, two evidence tiers with a declared boundary between mixing and porting, and four falsifiability conditions on the typology (§1.2, §9).
2. **A four-channel application of the apparatus**: formal, controlled, quantitative, narrative — each channel an anchored application whose claims are stated at the supply level its

checks support (§2–§5).

3. **A mapping (M1–M5) between the "hardness" criteria of hard science fiction and the falsifiability criteria of AI research claims**, with a four-stage pipeline that ports the former into the latter (§6).
4. **One engineering-grade, mid-strength instance of the full pipeline**, on record and examined in §7 on the same footing as the historical cases, scored by the same criteria, with its negative results disclosed in full.
5. **An adjudication agenda for the reverse channel**: what would count as machine worldviews feeding back into science, stated so that it can fail (§8).

The order is itself a graded claim: the apparatus (1) is what this paper would be missed for; the channel readings (2) are its first application and are graded at the supply level by design, and the paper asks to be credited with the checks that produced the grades, not with the sweep of the channels.

Four scope declarations bound what follows. (i) This is a position paper with an intellectual-historical core; its historical sections are case studies, not a systematic history. (ii) The reference pool is closed: fifty-six items, cited by pool number [R01]–[R56]; pool numbers count entries, not works — [R12] carries two primary texts under one entry because they anchor the same lexical item. Fifty-five entries were verified against primary records; one ([R56]) is the authors' own companion study, anonymized in this edition (title, authors, and identifiers withheld, to be restored in a revised version, available to any reader from the author on request) and attested by its authors rather than independently verified. Batch-by-batch verification records for the pool are consolidated in the Appendix. (iii) The historical evidence tier and the pre-registered adjudication tier are distinct instruments (§1.2); nothing in the historical sections borrows the authority of the latter. (iv) Uniqueness-flavored statements — including the statement that no integrated treatment of the four channels exists — are made strictly relative to our three-line search, completed 2026-09-16; its protocol and limits are stated in the Appendix and inherited by every statement made relative to it. The closest dialogue partner that search returned [R42] works a comparative taxonomy of fiction's roles; it carries neither a four-channel typology with scored cases nor an adjudication mechanism, so the two projects are complementary rather than competing.

A note on numbers: every number in this paper is bibliographic, a formula named in the text, or an experimental number cited by field path into the frozen record examined in §7. The term "machine worldview" appears as in the thesis sentence; its definitional anchor is cited in §8.

2 Channel I: Formal Fiction into Physics

2.1 The historical chain

A geometry can be built before there is anything for it to describe. The nineteenth century produced consistent alternatives to Euclidean geometry as pure formal constructions: at the time of their construction no physical theory made use of them. The physical speculation is quoted and dated. In the habilitation lecture, Riemann makes the properties distinguishing space a matter "only to be deduced from experience," seeks the ground of spatial metric relations outside space, "in binding forces which act upon it," and hands the open end to "the domain of another science, of physic" [R53]. Clifford proposed that the variation of the curvature of space is "what really happens in that phenomenon which we call the motion of matter" [R54]. Helmholtz made the public case for the empirical standing of geometrical axioms: on the empirical hypothesis, axioms "can be inferred from facts of experience and so established, or if, when tested by experience, they are perhaps to be rejected"; measurements that "confirm empirically the axiom of parallels" leave the curvature of our space "thus far to be indistinguishable from zero" [R55].

Yet this remained speculation, never an adopted physical theory: they were languages in search of a subject matter. We use "formal fiction" in exactly this sense — a self-consistent symbolic system not adopted, and not coordinated, as the language of any physical theory at the time of its construction — and not as a rhetorical flourish. The constructors' speculation is recorded, not erased, but no physical theory took it up, and that is the distinction the definition draws; it makes the channel's input state checkable by a datable historical question.

How one such language was taken up by physics is documented in detail by Walter [R01]: Minkowski's four-dimensional reformulation of special relativity carried a recognizably non-Euclidean style, and the style was itself part of what was transmitted — not the mathematics alone, but a way of writing, teaching, and receiving relativity, treated by Walter as a historiographical object in its own right. The chain continues through general relativity, whose formulation took Riemannian geometry as its working language. By then the direction of travel was fixed: a formal apparatus originally developed with no physical referent had become the standard medium in which a physical theory was stated; the geometry did not need the physics to be consistent, and the physics arrived later and found the language available.

That order — the language precedes the empirical joining — is the signature of this channel, and unlike most signatures in intellectual history it is anchorable in primary texts: the constructions, the reformulation, and the participant's account of the join all carry dates. Channel I is presented first as the template for what an anchored channel looks like, not as the strongest claim the paper makes.

2.2 The join, in the participant's testimony

Einstein's 1921 lecture "Geometry and Experience", published the following year [R02], is the participant testimony for how the join was made. Einstein's account separates pure geometry — a formal system whose certainty holds precisely insofar as it abstains from reference to reality — from applied geometry, which acquires empirical content through a stipulated coordination between formal terms and physical situations. Certainty and reference are, on this account, a trade. In general relativity the trade is carried by the field laws together with the geometry: the coordination is an act performed inside physics, not a property waiting inside the fiction to be discovered.

For the channel this fixes the temporal order — the formal system precedes the coordination — and the division of labor: the fictional input supplies structure, the physical commitment is added later by the person best placed to know, with the join carrying a participant account. It also explains why the channel's residue is a language rather than a fact: the empirical content lives in the coordination, not in the geometry.

Wigner's "unreasonable effectiveness of mathematics in the natural sciences" [R03] names the residual surprise of the pattern; we use it as the channel's counter-question, not its explanation — it records that something is unexplained after the anchors are laid, and the channel claim below spends no more than the anchors support.

2.3 Channel claim and assessment

Channel claim. Formal fiction supplies self-consistent language prior to reference, and the feedback occurs as language preceding the empirical joining.

Channel assessment.

- *T (temporal priority).* The formal structures precede their physical deployment: the constructions predate the uses, the transmission of the non-Euclidean style into physics is documented in the scholarly record [R01], and the join is fixed by the participant's own account [R02]. Every link is primary-text anchorable. **Pass.**

- *D (dispensability).* Here the channel is deliberately graded down, and the scholarship says so: Walter's study shows the non-Euclidean style was not the unique path into relativity — one presentation among available alternatives, adopted for reasons that include notation and pedagogy as much as necessity [R01]. The check therefore downgrades the claim from necessity to supply: the fictional language was available, adopted, and consequential — not indispensable. This is the check doing its job: it prevents the channel claim from drifting into a necessity thesis. **Partial pass; the claim is stated at the supply level.**
- *R (operative residue).* The residue is a formal-geometric language that remains the standard medium of the theory — an operative residue of exactly the kind that the temporal anchor and the supply-level downgrade leave in place. **Pass.**

The asymmetry in the scores is informative rather than embarrassing. Walter's non-uniqueness evidence [R01] sets the internal tier: the language supply is strongly documented, the necessity of the style is not claimed, and "language prior to reference" is exactly the claim that tier supports.

3 Channel II: Controlled Fiction

3.1 The thought-experiment dispute

Thought experiments are the canonical case of fiction doing scientific work, and the philosophical literature contains a live dispute over how the work is done. On the realist side, Brown [R04] holds that in the stock cases, thought experiments yield new knowledge of laws of nature not reducible to derivation: the scenario, once constructed, shows something that running through premises does not by itself deliver. On the eliminativist side, Norton [R05] holds that a thought experiment is an argument in picturesque clothing: its epistemic work is fully done by a reconstructible argument from premises to conclusion, and the imagery is a vehicle, not an independent source of warrant. The dispute is live, the positions are incompatible as accounts of mechanism, and this paper does not adjudicate them; for the current state of the wider literature, see the recent survey [R45]. The case is canonical for this paper for a structural reason: the fictional status of the input is not in dispute — the scenario describes an experiment that was not run — so the dispute isolates exactly the question the channel faces, what does the fiction contribute and how would we know, with none of the fog that surrounds weaker senses of "fiction."

What matters for the channel is a point the two sides share: whatever the yield, both parties hold it can be exhibited in a checkable form — Brown by isolating the insight, Norton by reconstructing the argument. The disagreement is over where the epistemic weight sits. Translatability is therefore the common ground of the dispute, and exactly the property the channel criterion needs: the channel runs on either side's terms, because either side can produce the checkable artifact.

3.2 Model-based reasoning and fictionalism

Nersessian [R06] places the thought experiment inside a wider class of model-based reasoning — concepts created through iterative construction of imagistic, analogical, and extreme-case models — and Suárez's collection [R07] supplies the fictionalist frame: fictions can represent without describing anything real, and their success does not presuppose a corresponding real property. Both lines move the question from "is the fiction true?" to "what work does the fiction do, and how is the work checked?" — the question our criteria operationalize; and they widen the channel's supply: idealized scenarios carry representational load in concept formation [R06] and modeling [R07] generally. For the assessment below, Norton's reconstruction program

is the dispensability check in its strongest form, and the fictionalist frame explains how a fiction can be a working input while remaining a fiction.

3.3 The operational definition of "controlled"

We call a fictional construction **controlled** when its epistemic yield is translatable into a checkable argument or model. The channel's criterion takes the broader layer, content-translatability — the yield, once laid out, can be checked by a competent third party; warrant-translatability (re-derivable from the reconstruction alone) is the stricter form, imposed by §8's adjudication agenda on AI-generated hypotheses, which is one reason the two ends of the loop share criteria.

Two boundary clarifications: translatability is a property of the yield, not the imagery — the criterion does not require the dressing to be disposable, and so takes no sides in §3.1's dispute; and it is a threshold, not a quality grade — a yield that cannot be laid out as argument or model is outside the channel's scope, however inspiring. Translatability is thus the boundary between controlled fiction and free imagination, the channel's criterion rather than a courtesy: Norton's reconstructions are precisely translations, and Brown's stock cases are scenarios whose yields survive translation.

Channel claim. Controlled fiction is a routine production material of science, and the channel criterion is translatability.

Channel assessment.

- *T (temporal priority).* The scenario is constructed before the argument it organizes: construction is the fictional step, reconstruction the checkable one, and in the stock cases of the literature the construction is datable and the reconstruction follows it [R04], [R05]. **Pass.**
- *D (dispensability).* This is the disputed check by construction: Norton's position is that the fiction is dispensable, Brown's that it is not. The channel claim does not need the dispute settled. What it claims is production value under content-translation, not indispensability of the imagery; and if Norton is right — warrant-translatability being the stricter form the content layer can take — the channel's residue is the translated argument itself, which still passes R. **Unresolved by design; the claim is worded to be invariant under either resolution.**
- *R (operative residue).* The residue is a reusable argumentative and modeling technology — limiting cases, idealizations, reconstruction templates — that remains in routine scientific use after any particular scenario has served. **Pass.**

The channel passes T and R, with D unresolved by design. The invariance of the claim under the Brown–Norton outcome is a feature of the criterion, not a dodge: translatability was chosen because both disputants grant it.

4 Channel III: An Abstract Quantity Becomes an Engineering Quantity

4.1 Entropy's two journeys

Channel III (the quantitative channel) runs on quantities, and its input is an abstraction rather than a fiction in the story sense. A quantity defined over an abstract, deliberately idealized object — an ensemble of messages with no material location — can travel into engineering by formal analogy and travel back into physics carrying a measurable price. The channel is classified accordingly as abstract-quantity/formal-analogy rather than fiction proper, and one fact settles the classification: statistical mechanics had given entropy a statistical reading over microstates long before the quantity was defined over message ensembles, so the physical quantity is the elder of the two and the channel cannot claim that the abstract preceded the physical. What

the channel claims, and what its dated arrows carry, is the journey of the defined quantity itself — from definition over an abstract object, into engineering, and back into physics at a price.

The stock example is entropy's second life. Statistical mechanics had given entropy a statistical reading over microstates from its nineteenth-century foundations — roughly seventy years before the information-theoretic definition, a precedence the T score below is graded against rather than around. Shannon [R09] then defined the information produced by a message source as $H = -\Sigma\ p \cdot \log\ p$ — a quantity whose referent is an ensemble of messages, an object with no mass, no location, and no temperature. The definitional act is the abstracting step, and it is datable. Jaynes [R08] built the bridge back toward physics: recasting statistical mechanics as inference, he showed that the same maximization principle that governs the informational quantity governs the derivation of physical distributions — the identity of form between the two quantities was made to carry weight rather than to decorate. Landauer [R10] completed the circuit by rematerializing the quantity: the erasure of information, he showed, carries a thermodynamic cost of $kT \cdot \ln 2$ per bit, so the abstract measure re-entered physics as an expense with experimental consequences in principle — a price that a physical apparatus pays for a bookkeeping operation on an abstract object.

Each arrow in this circuit is anchored to a datable primary source: the definitional act to Shannon [R09], the bridge to Jaynes [R08], the rematerialization to Landauer [R10]; no arrow rests on recollection, and the circuit is made of definitions, derivations, and a cost, all checkable in the primary texts. Each arrow also stands without the others — which is why the verdict below is a conjunction of separately anchored facts.

4.2 Verdict and channel claim

The channel's verdict rests on three observable facts. The isomorphism of formula: the same functional form — $H = -\Sigma\ p \cdot \log\ p$ — up to a positive multiplicative constant (Boltzmann's k, with its physical units) and the choice of logarithm base, governs the abstract ensemble and the physical distribution — not a similar expression, the same one up to exactly those factors. The rematerialization: the quantity acquires an experimental price in principle, $kT \cdot \ln 2$ per erased bit [R10] , now traced across survey and dynamical treatments of information thermodynamics [R43], [R44]. And the standing of information in physical reasoning today: a quantity with thermodynamic consequences rather than a metaphor.

Channel claim. An abstract quantity, defined over an idealized object by a datable act, can enter engineering by formal analogy and flow back into physics at a measurable cost; every arrow in the circuit has an anchor, and the channel claims no priority of the abstract over the physical.

Channel assessment.

- *T (temporal priority), scored against the real chronology.* Each arrow is dated by its primary publication [R08], [R09], [R10] — at the arrow level, the best-dated material in this paper. The channel-level demand — the input preceding the scientific use credited to it — does not survive the chronology: statistical mechanics had given entropy a statistical reading over microstates roughly seventy years before Shannon's definitional act, so a kindred physical quantity predates the abstraction. **Scoped pass: arrow-level dating holds; the channel-level priority claim is downgraded, and the channel is labeled abstract-quantity/formal-analogy accordingly.**
- *D (dispensability).* Whether communication theory would be equally parsimonious without H is decided by nothing in the pool, and we do not assert it. The channel claim rests on the anchorability of the arrows, not the indispensability of the quantity — the same supply-level discipline as Channel I. **Indeterminate; graded accordingly.**
- *R (operative residue).* The residue is an operative quantity with units, prices, and experimental consequences in principle. **Pass.**

The channel holds T at the arrow level with the channel-level priority downgraded; D is indeterminate; the claim is stated at the anchored-arrow level. The difference from Channel I's "partial pass" is evidential, not rhetorical: there the pool carries positive evidence of non-uniqueness [R01] and the check runs, downgrading the claim to the supply level; here the pool contains no study of alternative formulations, so the check cannot run, and the claim is graded to be invariant under its outcome. This is the most mechanical of the four channels, deliberately so: quantities carry their definitions with them, so the feedback is exhibited as a sameness of functional form plus a cost — though not, as the T score records, without conceding that the physical quantity came first.

5 Channel IV: Narrative Fiction into the AI Agenda

5.1 A direct agenda-setting case

The cleanest case in this channel is also the most literal. Sagan's novel Contact [R40] turns on rapid interstellar travel; the novelist's inquiry to a physicist — whether such travel could be made scientifically respectable — is the documented hinge of the case [R41], and the answer became the published analysis of traversable wormholes by Morris and Thorne [R11]. The passability question was answered as physics: what holds a wormhole open, and what conditions a traveler must meet, are derived in the journal record, stated as equations rather than as plot.

The evidence structure is what makes the case flagship-grade for this channel. The chain from the novelist's question to the published solution is anchored by the authors' own accounts of the episode (**self-statement anchor**); the scientific content is anchored by the peer-reviewed paper itself (**primary-source anchor**). Narrative-channel evidence usually has one anchor or the other — a story that influenced someone, or a paper whose fictional origin is hearsay; here both ends of the same chain are anchored, and the two anchor types meet at a datable event. The case also displays the channel's characteristic evidence problem, and its solution: influence usually leaves recollections and little else, so the channel's standard is that a recollection must meet a published artifact — recollection supplies the fictional end, the artifact the scientific end. The case is one case: its strength is evidentiary, and it does not by itself establish a pattern.

5.2 Ontological vocabulary

Narrative fiction also supplies the nouns an agenda needs, and vocabulary supply is the channel's most mechanical evidence form, because dictionary records are checkable artifacts. The word robot entered the language through Karel Čapek's play R.U.R. (1920) — the word itself was suggested to Čapek by his brother Josef Čapek, as Čapek himself acknowledged; the Oxford English Dictionary's earliest recorded attestation of robotics falls in Asimov's 1941 story "Liar!" [R12]. These are **lexical anchors**: the record is public, dated, and independent of recollection. The vocabulary is not decorative: robot and robotics are the term of art for an entire artifact class across law, industry, and research, and the term descends, by dictionary record, from the fiction that named it.

The same holds for singularity. Vinge's 1993 essay [R13], published in an interdisciplinary NASA-sponsored volume, states the coming of intelligence beyond the human threshold as a near-term engineering question; the singularity vocabulary in its AI-agenda sense carries this named, datable textual anchor (**named textual anchor**). We make no claim about the word's earlier history; what the channel needs is that the agenda-bearing sense has an anchor, and it does.

A lexical or named textual anchor does double duty in the assessment below: it dates the vocabulary (T), and it exhibits the residue (R), since a term of art still in circulation is an operative residue by definition. Both anchors land in this paper's territory: the robotics

vocabulary names and regulates a class of autonomous artifacts, and the singularity vocabulary organizes the discourse about AI trajectories.

5.3 The quantitative tier

Beyond flagship cases and vocabulary, the channel has a quantitative tier: recent studies examine the association between science-fiction narratives and technology topics quantitatively [R14], [R15] (**quantitative-study tier**). The tier moves the channel from anecdote to measurement — the hypothesis that science fiction supplies topics to technology is testable at study level. We do not assert the strength or the direction of the association; the tier's contribution is its existence and form, and it disciplines the rest of the channel: the flagship case and the vocabulary are not asked to carry statistical weight, and the studies are not asked to explain single episodes.

5.4 Evidence tiers and the forward pointer

The channel's evidence comes in three grades, each with an anchor. Self-statement anchors: the participants document the fictional origin themselves ([R11]). Lexical and named textual anchors: dictionary records and dated primary texts fix the vocabulary ([R12], [R13]). Quantitative studies: the association is measured rather than recounted ([R14], [R15]). The grades are stated separately, and none is asked to carry more than it supports: the flagship case is one case; the vocabulary is supply, not necessity; the quantitative tier measures association, not mechanism. This grading is what allows the channel claim below to be exact about what narrative fiction does — it supplies nouns and it sets agendas — without borrowing strength from the grade above.

Channel claim. Narrative fiction both supplies vocabulary and sets agendas; the evidence is graded by anchor type, and every grade has an anchor.

Channel assessment.

- *T (temporal priority).* Precedences are datable, and each precedence carries its anchor type with it: the novel's question precedes the wormhole paper (self-statement anchor, [R11]); 1920 and 1941 precede the modern robotics vocabulary (lexical anchor, [R12]); 1993 precedes the discourse the singularity essay anchors (named textual anchor, [R13]). **Pass.**
- *D (dispensability).* Whether the same vocabulary and agenda would have arisen without the fictional sources is decided by nothing in the pool, and we do not assert it. Vocabulary supply is the kind of input that has substitutes; the channel claim does not need indispensability. **Indeterminate; graded accordingly.**
- *R (operative residue).* The residue is operative: a term of art in law, industry, and research; a discourse-organizing concept with a datable anchor; and a class of spacetime solutions entered into the physics literature with published conditions [R11]. **Pass.**

Channel IV is also where the loop stops being purely historical: an engineering-grade instance of the full pipeline — fictional world-building to formal specification to pre-registered adjudication to falsifiable experiment, with negative results disclosed in full — is on record and examined in §7 on the same footing as the cases of this paper; the checks transfer unchanged, and what changes is the tier on which the verdict products are graded (§1.2).

6 Method: Porting the Hard Criteria into AI Claim Adjudication

6.1 The criteria, taken from the genre's own definitions

Hard science fiction polices itself, and the policing has named criteria. The genre's reference entry quotes Allen Steele's working definition — "Hard sf is the form of imaginative literature that uses either established or carefully extrapolated science as its backbone" (Steele, "Hard Again", quote as cited in [R37]) — and states two guardrails in its own voice: "Hard sf should not, however, wilfully ignore or break known scientific principles", and hard sf "should seek to provide natural rather than supernatural or transcendental explanations for the events and phenomena it describes" [R37]. The same entry concedes that a rigorous definition of the category may be impossible [R37]. The anthologies that frame the modern hard-SF canon [R36] exemplify the tradition these statements describe; their capture limit is recorded in the Appendix. From this record, and strictly within it, this paper distills four working demands for "hard" — internal consistency, no supernatural checks, continuity with known science, and checkability of the world's claims by observers inside the world — and adopts them as the definition domain, inherited by the mapping in rows M1, M2, M3, and M5. Row M4 is declared rather than smuggled in: the genre's distilled record contains no freeze-before-the-run demand, so M4's criterion is imported from the pre-registration and third-party-recomputation literature ([R47]–[R50]) rather than mapped from the genre, and the diegetic-prototype record ([R32], [R33]) is carried in that row as motivation, not as the criterion's source. The reason for taking the genre's own standard rather than a philosophical ideal of rigor is strategic: the genre standard is operational — reviewers, readers, and writers apply it to manuscripts — and operational standards port; ideals of rigor do not.

Two preliminary reductions keep the mapping honest: the genre's criteria are criteria for a world, not a paper, so each must be re-expressed as a criterion for a claim; and the genre enforces through review practice, not adjudication protocols, so the mapping must supply the enforcement mechanism the genre never needed. Both appear as the mapping invariant below.

6.2 The mapping table

The mapping's invariant is a single question, identical on both ends of every row: given the ledger, can a third party recompute the conclusion from the ledger alone? On the hard-SF side the ledger is the world's stated physics; on the AI side it is the claim's stated support. Each row below pairs a genre criterion with its AI-research counterpart and names the failure mode the row exists to catch.

The table claims a constructed structural parallel — row-wise pairing of criteria with shared failure modes — and nothing stronger; "isomorphism" is reserved for the formula-level identity of Channel III, and no row-to-row isomorphism is claimed here. One row is an import, declared as such (M4; §6.1). It is not an equivalence theorem: the two sides differ in enforcement, and the differences are inherited by the pipeline. What makes each row a mapping rather than an analogy is the shared failure mode: a genre manuscript and an AI paper can fail the same row in the same way, and a third party can check for the failure in both.

Anonymous interface. The practice that the M4 row formalizes — freeze the adjudication before the run, and hold the record to a recomputability standard — is systematized in an archived companion study of auditability and falsifiability in LLM-reasoning research, cited here as the anonymized pool entry [R56]: withheld in this edition, to be restored in a revised version, available to any reader from the author on request, and attested by the authors of this paper rather than independently verified (§1.3(ii)). The row itself is this paper's formalization, and its criteria stand on the open literature ([R47]–[R50]) with or without that entry.

No.	Hard-SF criterion (pool source)	AI-claim criterion (provenance stated per row)	Shared failure mode
M1	Internal consistency: the world's books run no deficit — this paper's formalization, prompted by the guardrails recorded in [R37]	Claim and evidence structure agree; the support covers what is asserted — this paper's formalization (claim–support correspondence: [R46]), prompted by the failure modes documented in [R38]	Self-contradiction
M2	No supernatural checks are written — grounded in the entry's standard: "natural rather than supernatural or transcendental explanations for the events and phenomena it describes" (quote as cited in [R37])	No appeals to undelivered future capability [R39]	Deus ex machina
M3	Continuity with known science — grounded in the entry's guardrail: "Hard sf should not, however, wilfully ignore or break known scientific principles" (quote as cited in [R37])	Anchoring in existing reviewable artifacts — this paper's formalization, prompted by the failure modes documented in [R38]	Bare assertion
M4	No counterpart among the genre's four distilled demands: the record demands no freeze-before-the-run adjudication of a world, so this row's criterion is imported, not mapped — declared in §6.1. Motivation, not source: narrative artifacts can carry engineering commitments; the diegetic prototype has documented instances [R32], [R33]	Adjudication procedure pre-registered, recomputable by a third party — this paper's formalization (pre-registration: [R47], [R48]; third-party recomputation: [R49], [R50]), an import per the left cell, prompted by the failure modes documented in [R38], [R39]	Changing the rules after the fact
M5	The world's claims are checkable by observers inside the world — this paper's distillation, grounded in the definition the entry quotes: "established or carefully extrapolated science as its backbone" (Steele, quote as cited in [R37])	Falsifiable, with the kill line defined before the experiment runs — this paper's formalization (falsifiability: [R51], [R52]), prompted by the failure modes documented in [R38], [R39]	Unfalsifiable rhetoric

Two sources carry the failure-mode side of the table: Wagstaff's method criticism, which argues that machine learning research should be judged by delivered, real-world capability rather than by benchmark claims alone [R38], and Lipton and Steinhardt's catalogue of troubling trends in which rhetoric outruns evidence [R39]. Of the AI-side criteria, only M2's no-future-capability demand restates a concern one of these sources states directly; the criteria of M1, M3, M4, and M5 are this paper's formalization, prompted by the failure modes those sources document, with their methodological originals cited in place ([R46]–[R52]; row-by-row provenance consolidated in the Appendix). The originals supply the practices the criteria formalize; the formalization, and the mapping, remain this paper's. The failure modes are those two sources'; the criteria that catch them are this paper's.

6.3 What exists, and where it stops

The practice of using narrative artifacts to steer real research and development is documented, not hypothetical, and the pool carries six entries across three literatures: science fiction prototyping proposes fictional scenarios as a source of technology ideas for engineering audiences [R30]; design fiction states the program in its general form [R31] and operates it as an empirical HCI method [R35]; sociological studies of the film industry document cinematic depictions feeding back into laboratory practice [R32], [R33]; and an analysis of computing's future discourse shows research agendas organized by the imaginaries they inherit [R34]. Together these establish the forward channel's practical existence at the level of agendas and prototypes.

What this line does not have is an adjudication tier: its methods are generative and ethnographic — they produce ideas and document influence, but they do not freeze a verdict function in advance, and they do not kill. The stop is not a failure of the literature; it is where its questions end. On the other side, the ML method-criticism literature names the failure modes — unsupported claims, post-hoc rules, rhetoric outrunning evidence [R38], [R39] — but no pipeline that ports a genre's own discipline into claim adjudication. The two literatures face each other across exactly the gap the mapping table formalizes.

To the extent of the three-line search recorded in §1.3(iv), the mapping has not been stated as a structural parallel with a portable pipeline.

6.4 The four-stage pipeline

The pipeline has four stages, each with an input, a product, a criterion from the table, and a named failure mode.

- **S (Setting).** Input: narrative fiction. Product: a self-consistent world setting with its own ontological vocabulary. Criterion: M1. Failure mode: a hidden contradiction in the world's books.
- **F (Formalization).** Input: the setting. Product: a formal system — identities, state spaces, metrics — into which the setting's vocabulary is transcribed. Criteria: M2, M3. Failure mode: metaphor with no formal counterpart.
- **P (Pre-registration).** Input: the formal system. Product: a frozen adjudication function with an external anchor, fixed before any run. Criterion: M4. Failure mode: changing the rules after seeing the data.
- **E (Experiment).** Input: the frozen adjudication function. Product: a verdict — including a kill. Criterion: M5. Failure mode: unfalsifiable rhetoric.

The pipeline's discipline is concentrated in one requirement: the kill line must be defined before the experiment runs, and the verdict function must be executable by a third party who did not build the system. A pipeline that can merely confirm is a pipeline with M5 removed, and M5 is the row that makes the rest of the mapping more than rhetoric. One calibration belongs with the requirement rather than in a footnote: executability is a property of the record,

not a report of who has executed. The instance of §7 was built and run by its own authors; in this edition, third-party executability runs through the full digests available from the authors on request (§7.1), public executability is deferred to a revised version, and a third party has in fact recomputed the anchors and headline numbers (§7.2). The requirement is therefore only partially met in this edition.

Method claim. The hard-SF criteria and the falsifiability criteria of AI research claims are structurally parallel under the mapping M1–M5 — row-wise pairing of criteria with shared failure modes, one row (M4) importing its criterion from the pre-registration literature and declared as such — and the four-stage pipeline ports the mapping into a worked method, demonstrated on one artifact on the given benchmarks at the stated statistical power (§7.3), not as a general working methodology. The claim weakens if any row's two ends fail to share form and failure mode under inspection, or if the pipeline cannot be executed end to end on a concrete artifact. Section 7 reports such an execution.

7 Case Study: An Anonymized Engineering Instance

This section examines one documented artifact — an engineering project whose identity is anonymized in this edition and restored in a revised version — on the same footing as the cases of §2–§5. The anonymization covers the project's name, its fictitious entity's name, and the project-name tokens inside artifact basenames (replaced below by the marker "anon"); full names, path prefixes, and the public pointer are withheld in this edition and available from the authors on request. The token "GT" that remains inside some artifact basenames below is the field-generic abbreviation of "ground truth," not a project-name token; the basenames are part of what the frozen digests cover and cannot be altered without breaking the anchors they attest. It touches no anchor digest, no number, and no field-path semantics: the five anchor prefixes, every measured value, and every field path in this section are unchanged by it. The tier discipline of §1.2 applies in full: what follows is the pre-registered adjudication tier, not the historical tier, and every number is cited by its field path into the frozen record.

7.1 The chain, replayed end to end

S. A science-fiction worldbuilding exercise posits a fictitious scattering entity (anonymized here) whose physics is a three-channel energy law: an incident path's energy splits into transmission, reflection, and irreversible dissipation. The setting supplies a conservation bookkeeping identity and a vocabulary for reasoning about exchange.

F. The formalization binds each node of an LLM-generated concept-decomposition graph to a two-parameter state of that entity, so that traversing reasoning paths undergo three-channel scattering obeying the constructive identity T+R+A=1 for arbitrary parameters; the reverse dynamics are additionally formalized as a game on the graph, with an audited scalar carrying monotonicity and near-gradient readings, and the empirical coordination ratio (ECR) as the distribution-level metric.

P. The adjudication layer is frozen before runs as pure verdict functions with SHA-256 anchors, printed here as twelve-character prefixes and consolidated in Table 1 (frozen handoff artifact, field `preregistration_anchors_sha256_12`). Every key number below maps to a frozen field; hand-copying is forbidden by the artifact's own protocol.

E. Three results, in ascending order of severity.

The conservation identity survives audit. The maximum per-path deviation across all variants and benchmarks is $2.220446049250313 \times 10^{-16}$ — the scale of double-precision machine epsilon, against an implementation tolerance of 10^{-6} — closed (pre-registered). The quantified coordination reading lands at median ECR = 4/3 over the 17 finite-valued graphs, with coverage of 20 of 22 graphs; three graphs read Infinity, two graphs lack arm data, both disclosed as-is.

Frozen artifact	SHA-256 prefix
`GT_FORMALIZATION_v1.md`	aeefb8ef6972
`run_v21_gtformal.py`	9bbe43f41fa8
`run_v22_p1c.py`	6e9673205dc0
`SPEC_GT2B.md`	68a5b08ef007
`SPEC_GT8C.md`	6b09de9911c0

Table 1: The five frozen pre-registration anchors; prefixes are twelve-character SHA-256 values, byte-identical to the frozen record.

The dynamical tier dies. The kill protocol is systematic in its graph sampling (61 graphs in five families) and complete at the state level: all 6,760 states of the 338-task corpus, 20 steps each. All three formalized dynamical-equivalence propositions are falsified. P1a: maximum deviation 0.8569, an O(1) gap that does not vanish with step size. T-P1b: minimum directional cosine -1.0. T-P1c: killed on both lines — an 81-point grid over $\tau \in [0,4]$ reaches minimum cosine -1.0 at the best global τ, the per-state optimal $\tau*$ again reaches -1.0, and the per-state $\tau*$ median is 0. The potential-game reading is downgraded accordingly, to approximate (cyclic-graph median residual 0.669).

Field paths for every number above are consolidated in Table 2; the availability terms of the complete frozen repository and the full SHA-256 digests are stated in the Appendix. A reader working from the public text alone cannot execute the recomputation that M4 asks of a third party, and this paper does not claim otherwise; making the record and its checks available during evaluation is the standard the machine-learning reproducibility program set for its own venues [R49]. That demand is now half met: an independent party recomputed the anchors and the headline numbers on 2026-09-18 (§7.2), while public availability of the record remains deferred to a revised version.

Result	Frozen field path
Conservation deviation ($2.220446049250313 \times 10^{-16}$)	`results/anon_benchmark_v1_4_gsm8k.json` · `physics_audit.t_plus_r_plus_a_max_deviation`
Median ECR and coverage (4/3; 20/22)	frozen handoff artifact, field `headline_results.GT4_ECR`
Kill scope (61 graphs; 6,760 states)	frozen handoff artifact, field `headline_results.enumeration_scope`
P1a / T-P1b / T-P1c	frozen handoff artifact, field `headline_results.P1a_P1b_T_P1c`; source `results/anon_v22_p1c.json` · `verdict.min_cos_at_best_global_tau` and `verdict.min_cos_per_state_best_tau`; measured fields in `results/anon_v21_gtformal.json` · `verdict.P1a_deviation.max_dev_inf` and `verdict.T_P1b.min_cosine`
Potential-game downgrade (0.669)	frozen handoff artifact, field `headline_results.P2`

Table 2: Field-path index for the E-stage results; every path is byte-identical to the frozen record.

7.2 Death is the point

The three falsifications are the pipeline working as designed, not a failure of it. M5 requires that the kill line be defined before the run; the run then killed. What survives at the dynamical level is no more than consistency-tier evidence, and the artifact's own verdict language records the

downgrade. Because verdict functions are frozen before their runs, the conservation pass and the dynamical kills stand at the same freeze level, and the pass cannot be read as the product of a lenient protocol. At the audit level, an independent rerun (2026-08-30) passed with bit-identical recomputation, the five anchors above, and nine provenance spot checks (frozen handoff artifact, field `headline_results.audit`) — the books were not merely kept but re-kept by a second pair of hands. A third party further recomputed the record on 2026-09-18: the five anchors matched bit-identically (5/5), and the headline numbers were traced item by item to the frozen fields (record: `_kimi_ftfb_s7_independent_recompute_2026_09_18.json`). Negative results are disclosed here as assets: what cannot be falsified cannot be killed.

7.3 What this instance is, and is not

The instance's honest positioning has three parts. (i) The channel attribute: the loop closes through the audit and epistemic channel, and the capability channel shows no detectable difference at the stated power. On real benchmarks the layer is indistinguishable from a trivial six-keyword rule filter reading nothing beyond label strings — on GSM8K the filter scores 0.87 against the layer's 0.85 (McNemar p=0.5), on StrategyQA 0.899 against 0.899 (p=1.0) — and at these sample sizes the detectable-difference threshold is roughly ±10 percentage points, so smaller increments are not excluded, disclosed as-is. Plain chain-of-thought significantly outperforms the layer on GSM8K, 0.97 versus 0.85. Fusion with a semantic prior dilutes rather than helps: physics 0.484→0.452 and historical 0.783→0.739 at the λ=0.5 convex combination. None of these numbers is massaged out of the record; field paths are consolidated in Table 3.

Result	**Frozen field path**
Filter comparison and power honesty (0.87/0.85, p=0.5; 0.899/0.899, p=1.0; ±10pp)	frozen handoff artifact, field `headline_results.E9_5_power_honest`; artifact `results/anon_v22_e95ci.json`
CoT baseline versus layer (0.97/0.85)	`results/anon_benchmark_v1_4_gsm8k.json` · `cot_baseline_accuracy` and `unified_accuracy`
Fusion at λ=0.5 (0.484→0.452; 0.783→0.739)	`results/anon_v20_crossval.json` · per-graph fields `prior_arm_eval.`<graph>`.hybrid_norm`@0.5

Table 3: Field-path index for the §7.3 results; every path is byte-identical to the frozen record.

(ii) The magnitude attribute. The instance runs structurally parallel to the Channel I pattern — a fictional physics supplies a formal language, the language is adjudicated, a verdict lands — but its magnitude is not comparable to the non-Euclidean case, and nothing in this paper claims otherwise. (iii) The not-yet attribute. A non-Euclidean-grade loop would require third-party adoption of the formal language by a community that did not build it; that adoption has not occurred, and no uniqueness claim is made about the instance — the positioning is exactly "engineering-grade, mid-strength."

7.4 Scoring on the same table

Scored by the three checks of §1.2, on the same footing as §2–§5; the consolidated table is §9.1. Temporal priority: the fictional setting precedes the formal system it seeds, anchored by the frozen record — an author-attested freeze: a digest attests that the artifact was not altered after the fact, but from inside the record it cannot, to a party that did not build the system, attest that the freeze preceded the runs; the anchor buys verifiability, not independence, and its standing is stated in the Appendix. The five printed anchor prefixes date the formalization and the verdict functions to before the runs they adjudicate, on the artifact's own attestation;

the freeze is now backed by an independent third-party recomputation (2026-09-18), but not yet by a trusted time-stamp or a third-party archive — the upgrade path is half executed. Dispensability: not claimed and not needed; the case rests on executability, and D is out of scope per §1.2. Operative residue: the machine-checkable ledger — identities, verdict functions, and anchors that continue to run after the fiction has done its work. The instance is therefore scored T as a provisional pass — the anchor is author-attested, of lesser independence than the historical rows — and R as a pass, on the pre-registered tier, with D out of scope; its evidential weight is one mid-strength case, asked to carry nothing more.

8 Outlook: Machine Worldviews and the Reverse Channel

Scope declaration. This section is an open-question agenda: it makes no prediction; its product is the set of conditions under which the reverse channel would become decidable — not evidence that it has opened.

8.1 The term, and the existence line

The term *machine worldview* has no settled definition. The survey literature covers the definitional debate over world models without closing it [R20], and this paper adds no definition of its own: the term enters here exactly as it entered the thesis sentence of §1.2. What this section needs is an existence line — that AI systems increasingly build and act through internal models of the domains they face — and that line is documented. Agents have been trained inside learned internal simulations of their environments since the work that opened this literature [R16]; world-model training has been scaled to master diverse domains in a single agent [R17]; generative systems have moved from producing frames to producing interactive environments [R18]; and a world-foundation-model platform has been proposed as shared infrastructure for physical AI [R19]. On the interpretability side, a sequence model trained on Othello moves — the game given as move sequences, never as rules — was found to carry an internal representation of the board [R21]: whatever "worldview" turns out to mean once the definitional debate settles [R20], an internal structure that encodes a domain the system was never told about is the phenomenon the term is reaching for. At the architectural level, a position paper makes world models the core component of a proposed program for autonomous machine intelligence [R22].

One difference point must be declared: the survey that anchors the definitional debate [R20] does not ask the question this section asks — whether such internal structures feed back into the sciences. That is a gap this section opens, not a criticism of that survey.

8.2 The pipeline line

On the production side, a line of systems now automates recognizable parts of the research pipeline itself: The AI Scientist generates research ideas, writes code, runs experiments, and writes the manuscript [R23], continued in a second version [R24]; Coscientist demonstrates autonomous planning and execution of chemistry experiments [R25]; a multi-agent co-scientist generates and ranks research proposals at scale [R26]; and Agent Laboratory wires ideation, experimentation, and write-up into a single workflow [R27]. The ideas such systems propose have also been placed under human blind review at scale, with expert reviewers judging machine-proposed and human-proposed ideas under controlled conditions [R28].

The pipeline line matters here for one reason: the S and F stages of §6.4 — setting and formalization — are now reachable by machine actors. What the line does not yet show, at the granularity of the pool's documentation, is the P stage: a frozen adjudication function with an external anchor and an executable kill line, fixed before any run. And the failure-mode audit any such stage would inherit is already written: the known failure modes of ML-based science,

data leakage prominent among them, are catalogued and reproducibility-tested as corruptions of conclusions that leave a paper's own books looking clean [R29]; that catalogue is where an audit of machine-generated hypotheses would begin.

8.3 The adjudication agenda

Declaration sentence. Machine worldviews may be forming, and they may be good ones; neither observation is evidence of feedback. Whether machine worldviews feed back into the sciences is decidable: it stands or falls on whether machine-generated hypotheses can be frozen into formal specifications, adjudicated by functions pre-registered with external anchors, and killed or kept by experiments whose kill lines were defined before the runs — the same pipeline as §6.4, the same checks as §1.2, no lower standard.

The agenda has three conditions, inherited from the pipeline: the machine's output must be landable in a formal specification fixed before adjudication (freezable); the adjudication function must be pre-registered and recomputable by a third party from the record alone (anchored) — the stricter warrant-translatability layer, imposed because the pre-registered tier can afford it; and the verdict must be executable as a decisive test (killable). F-C3 is the section's falsifiability condition: if machine-generated hypotheses, run under pre-registered adjudication, systematically leave no kill line to hang, the reverse channel is declared dead.

What would count as a positive result is stated with equal care, so that the burden does not fall silently on the negative side. A machine-generated hypothesis would count as reverse-channel feedback when (i) it is traceable to the machine's internal model — an input this paper treats as fiction-analogous rather than as a fictional input: whether a learned world model counts as a fiction in the sense of §2–§5 is a definitional question this paper does not settle, and the agenda does not lean on the answer; (ii) it precedes the human verification it receives; (iii) it survives a pre-registered kill attempt; and (iv) it leaves an operative residue — a vocabulary, a quantity, or a formal system — inside a scientific community that did not build the machine. The four elements map onto the three checks of §1.2 and onto F-C3; no one of them is sufficient alone, and none is asserted to exist today.

The audit vocabulary for such an adjudication — auditable when a third party that holds the record can recompute the claim from the record alone — is systematized in the companion study [R56] (§6.2). For the §7 instance, that availability runs through the on-request channel declared in §7.1 until a revised version; whether the offer is taken up is outside the authors' control (§7.1).

It is worth noticing the asymmetry with the forward channels. Sections §2–§5 reconstructed feedback after the fact, from anchors left behind by participants; the reverse channel, if it opens, would be feedback whose evidence regime is chosen in advance. The criteria of this paper were built for the historical tier; their portability to the pre-registered tier is what §6 established and §7 exhibited. The reverse channel is thus the case where this paper's two instruments meet.

9 Conclusion: The Criteria Turned on Themselves

A paper that lives by criteria should be willing to die by them. This closing section scores the paper's cases on the paper's own checks (§9.1), turns the same checks on the paper's own thesis, states the limitations (§9.2), and re-records the kill lines (§9.3).

9.1 The master table

The per-section scores of §2–§5 and §7.4 are consolidated here; nothing is re-scored.

Three patterns in the table. T holds at the anchor level in every scored row — provisionally in the pipeline instance, which alone rests on an author-attested freeze — and in every row the anchor named is the anchor that carries the claim. One row carries a scoped downgrade:

Row (section)	T (temporal priority)	D (dispensability)	R (operative residue)
Channel I — formal (§2)	Pass (strong; primary texts [R01], [R02])	Partial pass — downgraded to the supply level ([R01])	Pass (strong)
Channel II — controlled (§3)	Pass	Unresolved by design ([R04], [R05])	Pass
Channel III — quantitative (§4)	Scoped pass — arrows publication-dated [R08]–[R10]; channel-level priority downgraded (statistical entropy predates the definition by roughly seventy years)	Indeterminate	Pass (strong)
Channel IV — narrative (§5)	Pass (tiered anchors [R11]–[R13])	Indeterminate	Pass
Pipeline instance (§7)	Provisional pass (author-attested freeze; pre-registered tier — §7.4)	Out of scope (no indispensability claim)	Pass
Reverse channel (§8)	Not applicable — §8 asserts no empirical claim; F-C3 is an agenda condition, not a pending verdict	Not applicable	Not applicable

Channel III's arrows are dated [R08]–[R10], but the channel-level priority of the abstract quantity over the physical fails, so the row is scored as a scoped pass and the channel is labeled abstract-quantity/formal-analogy (§4). D is where the honesty of this paper lives: one partial pass with an explicit downgrade to the supply level (Channel I, [R01]); one dispute left unresolved because the claim is invariant under either outcome (Channel II); two indeterminates, where the pool contains no study that could run the check (Channels III and IV); and one out-of-scope, where indispensability is not what the instance claims (§7) — four dispositions, one dictionary, stated in §1.2. R passes in every scored row: every channel leaves an operative residue — a formal language, a modeling technology, a priced quantity, a vocabulary and an agenda, a machine-checkable ledger. The last row is not a score: §8 makes no empirical claim, so its cells read not applicable rather than pending — F-C3 is a condition the agenda can fail, not a verdict on evidence in hand.

One limit of the table's bite must be stated plainly: F-C1 demands pairwise separability of the four forms, and each channel currently carries one flagship case plus one instance, so the trigger condition — some pair returning identical grade profiles across all scored flagship cases — is close to formally unmet simply because the specimens are few. This is a limit of the check's current bite, not a pass on separability; F-C4 keeps the typology revisable as cases accrue.

The paper on its own checks. The table above scores the paper's objects; the promise of the section's title — the checks on the paper itself — is kept here, scores reported as they come. *T*: the apparatus was assembled after the cases it scores, so the paper's own construction runs setting-first, not pre-registration-first; what was fixed before distribution is the kill-line list of §9.3, a genuine but narrow pre-commitment. *D*: a partial pass, downgraded to the supply level: the four channels could be re-described in neutral terms (unreferenced formal systems, idealized scenarios, defined quantities, narrative sources) and the checks would survive the re-description; what would not carry over is the apparatus — the checks, the tiers, the disposition dictionary, the mapping — which is why §1.3 orders the apparatus first. *R*: pending, by the check's own definition, since residue is what remains after the encounter and the paper's encounter with its readers has not happened. The paper's self-score is T partial, D partial, R pending. A paper about kill lines that withheld its own would fail M5; a paper that graded itself above what its checks return would fail them more quietly.

9.2 Limitations

Four limitations, stated plainly. Selection effect: the four channels are visible because their cases left anchors; feedback that left no record is invisible to this method, and the typology is a typology of the documented. The same effect reaches the scores: T and R pass in every scored row of §9.1 partly because the cases were selected for anchorability and residue, so those passes are in part a product of case selection rather than an independent finding; D, which the cases were not selected to pass, is where the selection pressure shows least. Scale: each channel carries a small number of flagship cases, and the pipeline has one engineering-grade instance; no prevalence claim is made anywhere in this paper. Magnitude: structural parallel to the non-Euclidean pattern does not make the engineering instance comparable to it in scale (§7.3). Tiers: the historical tier and the pre-registered tier are different instruments, and nothing in §2–§5 borrows the authority of §7's. Two boundary conditions close the list: the reference pool is closed at fifty-six items (§1.3(ii)); and the novelty statements rest on the single-engine, non-systematic retrieval recorded in §1.3(iv), whose protocol and limits are stated in the Appendix. The nearest neighbor that search returned [R42] is a dialogue partner rather than a competitor: it maps the roles fiction plays around technoscience, while this paper scores what fictional inputs do to science, under named checks.

9.3 What would weaken this paper

The falsifiability conditions are re-recorded here, in substance unchanged from §1.2, operational readings included. F-C1: if the four forms prove pairwise inseparable under the three checks — some pair returning identical grade profiles across all scored flagship cases — the typology weakens. F-C2: if a flagship case fails the temporal check — its earliest dated record not earlier than the dated scientific use credited to it — or cannot sustain even the supply-level claim, no primary record carrying an operative residue, its channel is downgraded. F-C3: if machine-generated hypotheses, run under pre-registered adjudication, systematically leave no kill line to hang — in every case of a sample fixed before the run — the reverse channel is declared dead. F-C4: if new cases cannot be classified without ambiguity into one of the four forms under the same three checks — independent classifiers disagreeing, or more than one admissible form persisting — the typology is ad hoc. The mapping of §6 carries its own condition: if any row's two ends fail to share form and failure mode on inspection, the method claim weakens. These are the paper's kill lines, recorded before any reader runs them — because a paper about kill lines that withheld its own would fail M5, the row this paper has least standing to fail.

AI disclosure. The drafting model is GLM, used for drafting and revision support under author instruction. The claims, the criteria, the mapping, and every number were specified by the authors. AI participation went beyond drafting and is stated in full: retrieval, matching, and the entry-by-entry verification of the reference pool ran with machine assistance; the anchor digests and the frozen numbers were re-checked by script; and the draft was examined by multiple independent rounds of machine review before this revision, with each review finding adjudicated by the authors and its resolution recorded in the revision record rather than silently absorbed. The authors directed all work, adjudicated all findings, and bear full responsibility for the result. Each number in §7 maps to a field path in the frozen artifact cited in place.

Appendix: Verification Statement

Verification records consolidated here, in one place, so that the body of the paper carries one pointer per site.

Pool verification, batch by batch. Fifty-five of the fifty-six pool entries were verified against primary records. The base pool [R01]–[R41] was verified entry by entry at the first verification pass (title and identifier against a primary record: DOI, arXiv, or venue page). [R13]

and [R14] initially carried unstable identifier snapshots and were anchored by primary-source descriptors; both identifiers were closed at the 2026-09-18 deep-revision pass (R13: JSTOR stable article URL, retrieved 2026-09-18; R14: full standard citation, Proc. Cambridge Philos. Soc. 2 (1864–1876, printed 1876):157–158), so the pool now closes at 55 verified + 1 anonymized ([R56]) = 56. [R42]–[R45], added at the second-round verification pass, carry that pass as their verification record; [R46]–[R52], added at the dual-review source-supplement pass, were verified against publisher-level records; [R53]–[R54], added at the R2 literature-verification pass, were verified against full-text captures retrieved 2026-09-17; [R55] (Helmholtz), added at the R2 full-revision pass, was verified against the full-text capture of the Atkinson translation retrieved 2026-09-17. [R56] is attested by its authors rather than independently verified.

Retrieval protocol and screening rule. The three-line search of §1.3(iv), completed 2026-09-16, ran sixty-four query groups on a single commercial web-search engine: a pool-item line locating the anchor entries (twenty-three title-level queries), an identifier line re-verifying them against DOI, arXiv, and venue records (fourteen queries), and an adjudication/novelty line covering case-provenance and adjacent-literature look-ups (twenty-seven queries); the raw retrieval snapshots (five results per query) are retained in the project's literature line. Screening rule: no candidate enters the pool without first-hand verification of its title and identifier against a primary record; an entry without a stable identifier is omitted rather than kept. The record is a single-engine, non-systematic sweep, not a multi-database survey; the uniqueness statements say what this search returned, not what the literature at large contains, and the record's limits are inherited by every statement made relative to it.

[R36] capture limit. The anthologies that frame the modern hard-SF canon [R36] exemplify the tradition the genre's reference entry describes. Their framing is used as paraphrase, not quotation: the anthology's full text was not captured in the verification pass, so [R36] carries no direct quotation anywhere in this paper, and the limit is flagged here rather than papered over.

Row-by-row provenance, stated in full. M1's claim–support correspondence has its argumentation-theoretic original in Toulmin's layout of claims, data, and warrants [R46] — a theory of argument rather than a methodology paper, mounted with that genre declared. M2's no-future-capability demand restates the speculation critique of [R39] in criterion form. The pre-registration clauses of M4 stand on the preregistration revolution [R47] and the Registered Reports model [R48]; the third-party recomputation clause stands on the NeurIPS reproducibility program [R49] and the cross-disciplinary policy statement on computational reproducibility [R50]; the falsifiability and kill-line clauses of M5 stand on Popper's demarcation criterion [R51] and its working application to machine-learning interpretability research [R52]. M3's anchoring criterion claims no pool original; it is this paper's formalization outright. The failure modes are those of [R38] and [R39]; the criteria that catch them are this paper's.

Availability terms. The complete frozen repository and the full SHA-256 digests — not only the twelve-character prefixes printed in §7 — are available from the authors on request in this edition; the public pointer is deferred to a revised version. Whether the request side takes the offer up is outside the authors' control and is not claimed to have occurred beyond the third-party recomputation of 2026-09-18 recorded in §7.2.

Anchor standing. The §7 instance's temporal anchor is an author-attested freeze: a digest attests that the artifact was not altered after the fact, but from inside the record it cannot, to a party that did not build the system, attest that the freeze preceded the runs; that ordering is recorded by the artifact and its protocol, the builders' own. The upgrade path — trusted time-stamp, third-party archive — is half executed: an independent third-party recomputation (2026-09-18) verified anchors and headline numbers, while time-stamping and archiving have not occurred. T remains a provisional pass accordingly (§7.4).

References

The reference list is the closed pool of fifty-six items, cited in the text by pool number [R01]–[R56]. Pool numbers count entries, not works: [R12] carries two primary texts under one entry. One entry, [R56], is the authors' companion study, anonymized in this edition (title, authors, and identifiers withheld; to be restored in a revised version; available to any reader from the author on request; attested by the authors rather than independently verified). Entries are transcribed at the granularity of the verification record: titles appear where the record carries them.

Pool R01–R15 (cited in §2–§5)

R01. S. Walter, "The Non-Euclidean Style of Minkowskian Relativity", in J. Gray (ed.), *The Symbolic Universe*, Oxford University Press, 1999.

R02. A. Einstein, "Geometry and Experience" (1921 lecture), in *Sidelights on Relativity*, Methuen, 1922.

R03. E. Wigner, Comm. Pure Appl. Math. 13(1):1–14, 1960, DOI:10.1002/cpa.3160130102.

R04. J.R. Brown, *The Laboratory of the Mind: Thought Experiments in the Natural Sciences*, Routledge, 1991.

R05. J.D. Norton, Can. J. Philos. 26(3):333–366, 1996, DOI:10.1080/00455091.1996.10717457.

R06. N.J. Nersessian, *Creating Scientific Concepts*, MIT Press, 2008.

R07. M. Suárez (ed.), *Fictions in Science: Philosophical Essays on Modeling and Idealization*, Routledge, 2009.

R08. E.T. Jaynes, Phys. Rev. 106:620, 1957, DOI:10.1103/PhysRev.106.620.

R09. C.E. Shannon, Bell Syst. Tech. J. 27:379–423, 1948, DOI:10.1002/j.1538-7305.1948.tb01338.x.

R10. R. Landauer, IBM J. Res. Dev. 5(3):183–191, 1961, DOI:10.1147/rd.53.0183.

R11. M.S. Morris & K.S. Thorne, Am. J. Phys. 56(5):395–412, 1988, DOI:10.1119/1.15620.

R12. K. Čapek, *R.U.R.*, 1920; I. Asimov, "Liar!", *Astounding Science Fiction*, 1941 (lexical anchor: "robotics", OED earliest recorded attestation).

R13. V. Vinge, "The Coming Technological Singularity: How to Survive in the Post-Human Era", in *Vision 21: Interdisciplinary Science and Engineering in the Era of Cyberspace*, NASA Conference Publication CP-10129, NASA Lewis Research Center, 1993. NTRS Document ID 19940022856 (accession 94N27359), `https://ntrs.nasa.gov/citations/19940022856`.

R14. H. Osawa, D. Miyamoto, S. Hase, R. Saijo, K. Fukuchi, Y. Miyake, "Visions of artificial intelligence and robots in science fiction: a computational analysis", Int. J. Soc. Robot. 14(10):2123–2133, 2022, DOI:10.1007/s12369-022-00876-z.

R15. F. Appio et al., Technovation 141:103172, 2025, DOI:10.1016/j.technovation.2025.103172.

Pool R16–R29 (for §8, Outlook)

R16. D. Ha & J. Schmidhuber, arXiv:1803.10122.

R17. T. Hafner et al., arXiv:2301.04104.

R18. J. Bruce et al., arXiv:2402.15391.

R19. NVIDIA, arXiv:2501.03575.

R20. J. Ding et al., arXiv:2411.14499 (ACM Comput. Surv., 2025).

R21. K. Li et al., arXiv:2210.13382 (ICLR 2023).

R22. Y. LeCun, "A Path Towards Autonomous Machine Intelligence", OpenReview id BZ5a1r-kVsf, 2022.

R23. C. Lu et al., arXiv:2408.06292.

R24. The AI Scientist-v2, arXiv:2504.08066.

R25. D.A. Boiko et al., Nature 624:570–578, 2023, DOI:10.1038/s41586-023-06792-0.

R26. J. Gottweis et al., arXiv:2502.18864 (v1 title "Towards an AI co-scientist"; current abstract-page title "Accelerating scientific discovery with Co-Scientist").

R27. S. Schmidgall et al., arXiv:2501.04227.

R28. C. Si, N. Yang, T. Hashimoto, arXiv:2409.04109.

R29. S. Kapoor & A. Narayanan, Patterns 4(9):100804, 2023, DOI:10.1016/j.patter.2023.100804.

Pool R30–R39 (for §6, Methods)

R30. B.D. Johnson, *Science Fiction Prototyping*, Morgan & Claypool, 2011.

R31. J. Bleecker, "Design Fiction: A short essay on design, science, fact and fiction", Near Future Laboratory, 2009.

R32. D. Kirby, Soc. Stud. Sci. 40(1):41–70, 2010.

R33. D. Kirby, *Lab Coats in Hollywood*, MIT Press, 2011.

R34. P. Dourish & G. Bell, *Divining a Digital Future*, MIT Press, 2011.

R35. M. Blythe, "Research through design fiction: Narrative in real and imaginary abstracts", CHI 2014.

R36. D.G. Hartwell & K. Cramer (eds.), *The Hard SF Renaissance*, Tor, 2002.

R37. P. Nicholls, "Hard SF", *The Encyclopedia of Science Fiction*, online edition, `sf-encyclopedia.com/entry/hard_sf`, accessed 2026-09-17 (entry updated 7 April 2025).

R38. K. Wagstaff, ICML 2012, arXiv:1206.4656.

R39. Z. Lipton & J. Steinhardt, arXiv:1807.03341 (reprinted in ACM Queue, 2019).

Pool R40–R41 (adjunct to §5, added at verification)

R40. Carl Sagan, *Contact*, Simon & Schuster, New York, 1985.

R41. Kip S. Thorne, *Black Holes & Time Warps*, W.W. Norton, 1994.

Pool R42–R45 (adjunct to §1, §3, §4, §9; added at second-round verification)

R42. A. Belsunces Gonçalves & W. Mehnert, "What fiction does within and outside technoscience: Science fiction, fictional technofutures, and sociotechnical fictions", TATuP 35(2):16–21, 2026, DOI:10.14512/tatup.7281 (open access).

R43. E. Bormashenko, "Landauer's Principle: Past, Present and Future", Entropy 27(4):437, 2025, DOI:10.3390/e27040437 (review).

R44. C.-Y. Hsieh, "Dynamical Landauer principle: Quantifying information transmission by thermodynamics", Phys. Rev. Lett. 134:050404, 2025, DOI:10.1103/PhysRevLett.134.050404, arXiv:2201.12110.

R45. N. Miščević, *Thought Experiments*, Springer Cham, 2022, DOI:10.1007/978-3-030-81082-5.

Pool R46–R52 (adjunct to §6, §7; added at the dual-review source-supplement pass, 2026-09-16)

R46. S.E. Toulmin, *The Uses of Argument*, Cambridge University Press, 1958 (updated edition 2003).

R47. B.A. Nosek, C.R. Ebersole, A.C. DeHaven, D.T. Mellor, "The preregistration revolution", *Proc. Natl. Acad. Sci. USA* 115(11):2600–2606, 2018, DOI:10.1073/pnas.1708274114.

R48. C.D. Chambers, L. Tzavella, "The past, present and future of Registered Reports", *Nature Human Behaviour* 6(1):29–42, 2022, DOI:10.1038/s41562-021-01193-7.

R49. J. Pineau, P. Vincent-Lamarre, K. Sinha, V. Larivière, A. Beygelzimer, F. d'Alché-Buc, E. Fox, H. Larochelle, "Improving Reproducibility in Machine Learning Research (A Report from the NeurIPS 2019 Reproducibility Program)", *J. Mach. Learn. Res.* 22(164):1–20, 2021, arXiv:2003.12206.

R50. V. Stodden, M. McNutt, D.H. Bailey, E. Deelman, Y. Gil, B. Hanson, M.A. Heroux, J.P.A. Ioannidis, M. Taufer, "Enhancing reproducibility for computational methods", *Science* 354(6317):1240–1241, 2016, DOI:10.1126/science.aah6168.

R51. K.R. Popper, *The Logic of Scientific Discovery*, Hutchinson & Co., London, 1959 (English edition of *Logik der Forschung*, Julius Springer, Vienna, 1934).

R52. M.L. Leavitt, A.S. Morcos, "Towards falsifiable interpretability research", arXiv:2010.12016, 2020, DOI:10.48550/arXiv.2010.12016 (ML Retrospectives, Surveys & Meta-Analyses Workshop, NeurIPS 2020; equal contribution).

Pool R53–R54 (adjunct to §2; added at the R2 literature-verification pass, 2026-09-17)

R53. B. Riemann, "On the Hypotheses which lie at the Bases of Geometry", habilitation lecture; English translation by W.K. Clifford, *Nature*, Vol. VIII, Nos. 183, 184, pp. 14–17, 36, 37 (D.R. Wilkins transcription), full text: `https://www.emis.de/classics/Riemann/WKCGeom.tex` (retrieved 2026-09-17); German original locator: `https://www.maths.tcd.ie/pub/HistMath/People/Riemann/Geom/` (title verified, text not captured).

R54. W.K. Clifford, "On the Space-Theory of Matter", Wikisource transcription, `https://en.wikisource.org/wiki/On_the_Space-Theory_of_Matter` (index page: "On the Space-Theory of Matter.djvu"; page categorized "1876 works") (retrieved 2026-09-17).

Pool R55–R56 (adjunct to §2, §6–§8; added at the R2 full-revision pass, 2026-09-18)

R55. H. Helmholtz, "On the Origin and Significance of Geometrical Axioms" (lecture to the Docenten Verein, Heidelberg, 1870), in *Popular Lectures on Scientific Subjects*, Second Series, trans. E. Atkinson, London: Longmans, Green, and Co., 1908; full text: Project Gutenberg eBook no. 77725 (released 2026-01-17), retrieved and verified 2026-09-17.

R56. Authors' companion study on auditability and falsifiability in LLM-reasoning research (archived; title, authors, and identifiers withheld in this edition, to be restored in a revised version; available to any reader from the author on request). [AUTHOR-ANONYMIZED ENTRY]